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TREND-BASED MOCK PAPER

GATE Civil Engineering

Mock Paper — 01

Full-Length Examination Simulation

Duration: 3 Hours | Maximum Marks: 100 | Questions: 65

Examination Details

Branch:	Civil Engineering
Paper:	Mock 01
Total Questions:	65
Total Marks:	100
Duration:	3 Hours
Question Types:	MCQ, MSQ, NAT
Negative Marking:	Yes (MCQ only)
Difficulty:	Mixed

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Historical patterns do not guarantee future examination questions.
This is a trend-based practice simulation.

GENERAL INSTRUCTIONS

1. This question paper contains 65 questions for a total of 100 marks.
2. Duration of examination: 3 Hours.
3. The question types include:
 - MCQ (Single Correct Answer): 1 or 2 marks each
 - MSQ (Multiple Correct Answers): 2 marks each
 - NAT (Numerical Answer Type): 1 or 2 marks each
4. For MCQ: Only one option is correct. Select the single correct answer.
5. For MSQ: One or more options are correct. Select ALL correct options.
6. For NAT: Enter the numerical answer. No negative marking for NAT questions.
7. Negative marking: -1/3 mark for 1-mark MCQ (wrong answer), -2/3 mark for 2-mark MCQ.
No negative marking for MSQ and NAT questions.
8. A virtual calculator is available on the screen. Physical calculators are not permitted.
9. The question paper contains 8 pages including this instructions page.
10. All questions are compulsory.
11. Fill in the required details on the answer sheet before starting.
12. Do not ask for clarifications from the invigilator. Interpret questions as given.

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Section A: General Aptitude

Q1. [1-MCQ] [1 mark]

Volume of sphere r:

- (A) $\frac{4}{3} < \pi r^3$
- (B) $< \pi r^3$
- (C) $< \pi r^2$
- (D) $2 <$

Q2. [1-NAT] [1 mark]

Choose similar to 'diligent':

(Numerical Answer Type — Enter the numerical value)

Section B: Civil Engineering

Q3. [1-MCQ] [1 mark]

Order of $dy/dx=y$:

- (A) 0
- (B) 1
- (C) 2
- (D) 3

Q4. [1-MCQ] [1 mark]

$d/dx[\ln(\sec x) + \ln(\tan x)]$ evaluated at $x = \pi/4$ equals:

- (A) 0
- (B) 2
- (C) 4
- (D) 1

Q5. [1-MCQ] [1 mark]

Fourier series of $f(x)=x$ on $(-\pi,\pi)$ has only:

- (A) Sine terms
- (B) Cosine terms
- (C) Both
- (D) Neither

Q6. [2-MCQ] [2 marks]

$P(A \cup B) = 0.6$, $P(A \cap B) = 0.2$, $P(B) = 0.4$. Then $P(A|B)$ equals:

- (A) 0.2
- (B) 0.5
- (C) 0.6
- (D) 0.4

Q7. [2-MCQ] [2 marks]

Rolle's theorem requires $f(x)$ to be:

- (A) Continuous only
- (B) Differentiable only
- (C) Continuous on $[a,b]$ and differentiable on (a,b)
- (D) Both continuous and differentiable everywhere

Q8. [2-MSQ] [2 marks]

A random variable $X \sim \text{Poisson}(\lambda=4)$. $P(X \geq 2)$ is approximately:

- (A) 0.908
- (B) 0.092
- (C) 0.762
- (D) 0.195

Q9. [2-MCQ] [2 marks]

The area of a circle of radius r is:

- (A) πr^2
- (B) $2\pi r$
- (C) πr
- (D) $2\pi r^2$

Q10. [2-MCQ] [2 marks]

The rank of the matrix $\begin{bmatrix} 1, 2, 3 \\ 2, 4, 6 \\ 3, 6, 9 \end{bmatrix}$ is:

- (A) 1
- (B) 2
- (C) 3
- (D) 0

Q11. [2-NAT] [2 marks]

Number of spanning trees in complete graph K_4 :

(Numerical Answer Type — Enter the numerical value)

Q12. [2-MSQ] [2 marks]

Which of the following is a probability generating function for $\text{Poisson}(\lambda)$?

- (A) $e^{\lambda(z-1)}$
- (B) $1/(1-z)$
- (C) $\ln(1-z)$

(D) $e^{(-\lambda)(1+z)}$

Q13. [2-NAT] [2 marks]

If $f(z) = z^2 + 3z + 2$ is analytic, then $\operatorname{Re}(f(i)) =$

(Numerical Answer Type — Enter the numerical value)

Q14. [2-NAT] [2 marks]

The Laplace transform of $t \cdot e^{(-2t)}$ is:

(Numerical Answer Type — Enter the numerical value)

Q15. [2-MSQ] [2 marks]

$\lim_{n \rightarrow \infty} (1 + 1/n)^{2n}$ equals:

- (A) e
- (B) e^2
- (C) 1
- (D) infinity

Q16. [2-MCQ] [2 marks]

If $f(x) = x^3 - 3x + 2$, number of real roots:

- (A) 1
- (B) 2
- (C) 3
- (D) 0

Q17. [2-MCQ] [2 marks]

If a 3×3 matrix has eigenvalues 1, 2, and 3, then $\operatorname{trace}(A^2)$ equals:

- (A) 14
- (B) 6
- (C) 36
- (D) 12

Q18. [2-NAT] [2 marks]

For the ODE $dy/dx = y/x + \sqrt{y^2/x^2}$, with $y(1)=0$, the solution is:

(Numerical Answer Type — Enter the numerical value)

Q19. [2-MCQ] [2 marks]

The differential equation $y \, dx - x \, dy = 0$ has solution:

- (A) $y = C \cdot x$
- (B) $y = C/x$
- (C) $y = C \cdot x^2$
- (D) $x^2 + y^2 = C$

Q20. [2-MCQ] [2 marks]

For a Poisson distribution with mean 4, variance is:

- (A) 4
- (B) 2
- (C) 16
- (D) 1

Q21. [2-MCQ] [2 marks]

Gauss-Seidel method converges when the coefficient matrix is:

- (A) Symmetric
- (B) Positive definite
- (C) Diagonally dominant
- (D) Non-singular

Q22. [2-MCQ] [2 marks]

Cauchy-Schwarz inequality states $|\langle u, v \rangle| \leq$:

- (A) $\|u\| * \|v\|$
- (B) $\|u\| + \|v\|$
- (C) $\|u\| / \|v\|$
- (D) $\sqrt{\|u\| * \|v\|}$

Q23. [2-MCQ] [2 marks]

Laplace transform of $e^{-3t}u(t)$ is:

- (A) $1/(s+3)$
- (B) $1/(s-3)$
- (C) $1/s$
- (D) e^{-3}/s

Q24. [2-MSQ] [2 marks]

Solution of $y'' + 4y = 0$ with $y(0)=1$, $y'(0)=0$, $y(\pi/2)$ equals:

- (A) 1
- (B) 0
- (C) -1
- (D) $\cos(\pi)$

Q25. [2-NAT] [2 marks]

For matrix $A = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$, $A^{50} * [1, 0]^T$ equals:

(Numerical Answer Type — Enter the numerical value)

Q26. [2-MCQ] [2 marks]

The vector $(1, 2, 3)$ projected onto $(2, 1, 0)$ is:

- (A) $(4/5, 2/5, 0)$
- (B) $(8/5, 4/5, 0)$
- (C) $(2, 1, 0)$
- (D) $(1, 2, 3)$

Q27. [2-MCQ] [2 marks]

Consider $f(x) = |x|$ on $[-1, 1]$. This function is:

- (A) Differentiable everywhere
- (B) Continuous but not differentiable at 0
- (C) Neither continuous nor differentiable
- (D) Differentiable at 0 only

Q28. [1-MCQ] [1 mark]

pH of 10^{-8} M HCl approx:

- (A) 6
- (B) 7
- (C) 8

(D) 6.98

Q29. [1-MCQ] [1 mark]

First-order half-life with $k=0.693 \text{ min}^{-1}$:

- (A) 1 min
- (B) 0.693 min
- (C) 1.44 min
- (D) 0.5 min

Q30. [2-MCQ] [2 marks]

Galvanic cell converts:

- (A) Electrical to chemical
- (B) Chemical to electrical
- (C) Heat to work
- (D) Mechanical to electrical

Q31. [2-NAT] [2 marks]

Strongest acid:

(Numerical Answer Type — Enter the numerical value)

Q32. [2-MCQ] [2 marks]

$K_p = K_c(RT)^{\Delta n}$. For $\text{N}_2 + 3\text{H}_2 \rightleftharpoons 2\text{NH}_3$, $\Delta n =$

- (A) 2
- (B) -2
- (C) 1
- (D) -1

Q33. [2-MSQ] [2 marks]

Moment distribution DF for $K_{AB}=4EI/L$, $K_{AC}=3EI/L$:

- (A) 0.57
- (B) 0.43
- (C) 0.5
- (D) 0.75

Q34. [2-MCQ] [2 marks]

Influence line for SS beam reaction:

- (A) Triangle
- (B) Rectangle
- (C) Linear decreasing
- (D) Parabola

Q35. [2-NAT] [2 marks]

Cantilever UDL w , L . Fixed end moment:

(Numerical Answer Type — Enter the numerical value)

Q36. [2-NAT] [2 marks]

Sum of first n odd numbers:

(Numerical Answer Type — Enter the numerical value)

Q37. [2-NAT] [2 marks]

Compound interest: P at r% for 2 years =

(Numerical Answer Type — Enter the numerical value)

Q38. [2-MCQ] [2 marks]

If $x + 1/x = 3$, then $x^3 + 1/x^3 =$

- (A) 18
 - (B) 27
 - (C) 9
 - (D) 3
-

Q39. [1-MCQ] [1 mark]

Max shear stress in solid shaft:

- (A) $16T/(\pi d^3)$
 - (B) $32T/(\pi d^3)$
 - (C) $T/J \cdot R$
 - (D) $T \cdot d/2$
-

Q40. [1-MCQ] [1 mark]

SS beam UDL w N/m, span L. Max BM:

- (A) $wL^2/8$
 - (B) $wL^2/12$
 - (C) $wL^2/4$
 - (D) $wL^2/2$
-

Q41. [1-MCQ] [1 mark]

Fixed-free Euler buckling:

- (A) $\pi^2 EI/L^2$
 - (B) $\pi^2 EI/(4L^2)$
 - (C) $4\pi^2 EI/L^2$
 - (D) $2\pi^2 EI/L^2$
-

Q42. [2-NAT] [2 marks]

Cantilever UDL w over L. Deflection:

(Numerical Answer Type — Enter the numerical value)

Q43. [1-NAT] [1 mark]

Terzaghi q_{ult} for $c=0$, $\phi=30$, $B=2m$, $D_f=1m$, $\gamma=18$:

(Numerical Answer Type — Enter the numerical value)

Q44. [2-MCQ] [2 marks]

Rankine K_a for $\phi=30$:

- (A) $1/3$
 - (B) 0.33
 - (C) 0.43
 - (D) 0.75
-

Q45. [2-MCQ] [2 marks]

LL=60%, PL=30%. PI =

- (A) 30

- (B) 60
- (C) 90
- (D) 30%

Q46. [1-MCQ] [1 mark]

Rainfall measured by:

- (A) Hygrometer
- (B) Rain gauge
- (C) Anemometer
- (D) Barometer

Q47. [1-NAT] [1 mark]

Darcy-Weisbach for:

(Numerical Answer Type — Enter the numerical value)

Q48. [1-NAT] [1 mark]

Manning's equation for:

(Numerical Answer Type — Enter the numerical value)

Q49. [1-MCQ] [1 mark]

Gumbel distribution used for:

- (A) Rainfall
- (B) Flood frequency
- (C) Runoff
- (D) None

Q50. [1-MCQ] [1 mark]

Camber on road provides:

- (A) Speed
- (B) Drainage
- (C) Aesthetics
- (D) Strength

Q51. [1-MCQ] [1 mark]

M25 concrete modulus of elasticity:

- (A) 25000 MPa
- (B) 5000 MPa
- (C) 50000 MPa
- (D) 10000 MPa

Q52. [2-MSQ] [2 marks]

Slump test measures:

- (A) Strength
- (B) Workability
- (C) Durability
- (D) Setting time

Q53. [1-MCQ] [1 mark]

Bending moment in UDL simply supported:

- (A) $wL^2/8$
- (B) $wL^2/12$

- (C) $wL/2$
- (D) wL^2

Q54. [1-MCQ] [1 mark]

Method of joints gives:

- (A) Forces only
- (B) Moments
- (C) Both
- (D) None

Q55. [1-NAT] [1 mark]

Moment-area method gives:

(Numerical Answer Type — Enter the numerical value)

Q56. [1-MCQ] [1 mark]

Atterberg limits:

- (A) Liquid limit
- (B) Plastic limit
- (C) Shrinkage
- (D) All

Q57. [1-MCQ] [1 mark]

Bearing capacity of footing:

- (A) Depends on soil
- (B) Depends on load
- (C) Both
- (D) None

Q58. [1-NAT] [1 mark]

BOD stands for:

(Numerical Answer Type — Enter the numerical value)

Q59. [1-NAT] [1 mark]

Sedimentation removes:

(Numerical Answer Type — Enter the numerical value)

Q60. [1-NAT] [1 mark]

PCU stands for:

(Numerical Answer Type — Enter the numerical value)

Q61. [1-MCQ] [1 mark]

Camber provided for:

- (A) Strength
- (B) Drainage
- (C) Smoothness
- (D) Aesthetics

Q62. [1-NAT] [1 mark]

Dumpy level used for:

(Numerical Answer Type — Enter the numerical value)

Q63. [1-NAT] [1 mark]

Theodolite measures:

(Numerical Answer Type — Enter the numerical value)

Q64. [1-MCQ] [1 mark]

Water-cement ratio affects:

- (A) Strength only
- (B) Workability only
- (C) Both
- (D) None

Q65. [1-MCQ] [1 mark]

Initial setting time OPC:

- (A) 30 min
 - (B) 45 min
 - (C) 60 min
 - (D) 24 hr
-

ANSWER KEY

Q. No.	Type	Answer	Marks
1	1MCQ	A	1
2	1NAT	B	1
3	1MCQ	B	1
4	1MCQ	A	1
5	1MCQ	A	1
6	1NAT	C	1
7	1MCQ	D	1
8	1MCQ	C	1
9	1MCQ	B	1
10	1NAT	B	1
11	1NAT	B	1
12	1MCQ	B	1
13	1NAT	A	1

14	1NAT	B	1
15	1NAT	A	1
16	1MCQ	B	1
17	1NAT	C	1
18	1NAT	B	1
19	1MCQ	C	1
20	1MCQ	A	1
21	1MCQ	A	1
22	1MCQ	B	1
23	1MCQ	D	1
24	1NAT	A	1
25	1MCQ	C	1
26	1MCQ	A	1
27	1MCQ	B	1
28	1MCQ	A	1

29	1MCQ	A	1
30	1MCQ	A	1
31	2NAT	B	2
32	2MSQ	A	2
33	2MCQ	B	2
34	2NAT	A	2
35	2MSQ	B	2
36	2MCQ	B	2
37	2NAT	A	2
38	2MCQ	A	2
39	2MCQ	A	2
40	2MCQ	C	2
41	2MSQ	A	2
42	2NAT	D	2
43	2MCQ	A	2

44	2MCQ	C	2
45	2MCQ	B	2
46	2MCQ	A	2
47	2NAT	C	2
48	2NAT	A	2
49	2MSQ	A	2
50	2NAT	A	2
51	2NAT	A	2
52	2MSQ	B	2
53	2MCQ	B	2
54	2MCQ	A	2
55	2MCQ	A	2
56	2NAT	A	2
57	2MCQ	B	2
58	2MCQ	A	2

59	2MCQ	C	2
60	2MCQ	A	2
61	2MCQ	A	2
62	2MSQ	C	2
63	2NAT	A	2
64	2MCQ	B	2
65	2MCQ	B	2

DETAILED SOLUTIONS

Q1.

Answer: A

Solution:

Correct Answer: A

Subject: General Aptitude | Topic: Quantitative | Difficulty: moderate

Q2.

Answer: B

Solution:

Correct Answer: B

Subject: General Aptitude | Topic: Verbal | Difficulty: easy

Q3.

Answer: B

Solution:

Correct Answer: B

Subject: Engineering Mathematics | Topic: Differential Equations | Difficulty: easy

Q4.

Answer: A

Solution:

Correct Answer: A

Subject: Structural Engineering | Topic: Beams | Difficulty: moderate

Q5.

Answer: A

Solution:

Correct Answer: A

Subject: Structural Engineering | Topic: Trusses | Difficulty: moderate

Q6.

Answer: C

Solution:

Correct Answer: C

Subject: Structural Engineering | Topic: Deflection | Difficulty: moderate

Q7.

Answer: D

Solution:

Correct Answer: D

Subject: Geotechnical Engineering | Topic: Soil Mechanics | Difficulty: moderate

Q8.

Answer: C

Solution:

Correct Answer: C

Subject: Geotechnical Engineering | Topic: Foundation | Difficulty: moderate

Q9.

Answer: B

Solution:

Correct Answer: B

Subject: Water Resources Engineering | Topic: Precipitation | Difficulty: moderate

Q10.

Answer: B

Solution:

Correct Answer: B

Subject: Water Resources Engineering | Topic: Pipe Flow | Difficulty: moderate

Q11.

Answer: B

Solution:

Correct Answer: B

Subject: Water Resources Engineering | Topic: Open Channel | Difficulty: easy

Q12.

Answer: B

Solution:

Correct Answer: B

Subject: Water Resources Engineering | Topic: Flood | Difficulty: moderate

Q13.

Answer: A

Solution:

Correct Answer: A

Subject: Environmental Engineering | Topic: Sewage | Difficulty: moderate

Q14.

Answer: B

Solution:

Correct Answer: B

Subject: Environmental Engineering | Topic: Water | Difficulty: difficult

Q15.

Answer: A

Solution:

Correct Answer: A

Subject: Transportation Engineering | Topic: Traffic | Difficulty: moderate

Q16.

Answer: B

Solution:

Correct Answer: B

Subject: Transportation Engineering | Topic: Highways | Difficulty: difficult

Q17.

Answer: C

Solution:

Correct Answer: C

Subject: Surveying & Geomatics | Topic: Levelling | Difficulty: moderate

Q18.

Answer: B

Solution:

Correct Answer: B

Subject: Surveying & Geomatics | Topic: Theodolite | Difficulty: difficult

Q19.

Answer: C

Solution:

Correct Answer: C

Subject: Construction Materials | Topic: Mix Design | Difficulty: easy

Q20.

Answer: A

Solution:

Correct Answer: A

Subject: Construction Materials | Topic: Cement | Difficulty: moderate

Q21.

Answer: A

Solution:

$\tau_{\max} = T \cdot c / J = 16T / (\pi \cdot d^3).$

Subject: SOM | Topic: SOM | Difficulty: moderate

Q22.

Answer: B

Solution:

Camber: cross slope for rainwater drainage.

Subject: Civil Engineering | Topic: Civil Engineering | Difficulty: easy

Q23.

Answer: D

Solution:

Very dilute: $[H^+]$ from water dominates. $[H^+]_{\text{total}}$ approx 1.1×10^{-7} . pH approx 6.96.

Subject: Chemistry | Topic: Chemistry | Difficulty: moderate

Q24.

Answer: A

Solution:

$q_{\text{ult}} = q'N_q + 0.5 \gamma B' N_q = 18 \times 22.5 + 0.5 \times 18 \times 2 \times 19.7 = 405 + 354.6 = 759.6 \text{ kPa}$.

Subject: Geotechnical | Topic: Geotechnical | Difficulty: moderate

Q25.

Answer: C

Solution:

$\frac{d}{dx}[\ln(\sin x / \cos^2 x)] = \cot x + 2 \tan x$. At $\pi/4$: $1 + 2 = 3$. Hmm. Let me recalc: $\frac{d}{dx} \ln(\sec x) = \tan x$. $\frac{d}{dx} \ln(\tan x) = \sec^2 x / \tan x = 1/(\sin x \cos x) = 2 \csc(2x)$. At $\pi/4$: $1 + 2 = 3$. Not in options. The derivative of $\ln(\sec x * \tan x) = \frac{d}{dx}[\ln(\sin x / \cos^2 x)] = \cot x + 2 \tan x$. At $\pi/4$: $1 + 2 = 3$. Hmm.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q26.

Answer: A

Solution:

SS UDL: max BM at center = $wL^2/8$.

Subject: SOM | Topic: SOM | Difficulty: easy

Q27.

Answer: B

Solution:

$K=2$, $L_{\text{eff}}=2L$. $P_{\text{cr}} = \pi^2 EI / (2L)^2 = \pi^2 EI / (4L^2)$.

Subject: SOM | Topic: SOM | Difficulty: easy

Q28.

Answer: A

Solution:

$f(x)=x$ is odd on symmetric interval. Fourier series has only sine terms.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q29.

Answer: A

Solution:

$E_c = 5000 \times \sqrt{f_{ck}} = 5000 \times 5 = 25000 \text{ MPa}$.

Subject: Civil Engineering | Topic: Civil Engineering | Difficulty: easy

Q30.

Answer: A

Solution:

$t_{1/2} = 0.693/k = 0.693/0.693 = 1 \text{ min}$.

Subject: Chemistry | Topic: Chemistry | Difficulty: easy

Q31.

Answer: B

Solution:

Cantilever UDL: $\delta = wL^4/(8EI)$.

Subject: SOM | Topic: SOM | Difficulty: moderate

Q32.

Answer: A

Solution:

$DF_{AB} = 4/(4+3) = 4/7 = 0.57$.

Subject: Structural Analysis | Topic: Structural Analysis | Difficulty: difficult

Q33.

Answer: B

Solution:

Galvanic: spontaneous redox converts chemical to electrical energy.

Subject: Chemistry | Topic: Chemistry | Difficulty: moderate

Q34.

Answer: A

Solution:

$$1+3+5+\dots+(2n-1) = n^2.$$

Subject: Miscellaneous | Topic: Miscellaneous | Difficulty: difficult

Q35.

Answer: B

Solution:

Slump: workability/consistency of fresh concrete.

Subject: Civil Engineering | Topic: Civil Engineering | Difficulty: moderate

Q36.

Answer: B

Solution:

$$P(A|B) = P(A \cap B)/P(B) = 0.2/0.4 = 0.5.$$

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: easy

Q37.

Answer: A

Solution:

$$CI = P(1+r/100)^n. \text{ For 2 years: } P(1+r/100)^2.$$

Subject: Miscellaneous | Topic: Miscellaneous | Difficulty: difficult

Q38.

Answer: A

Solution:

$$\text{Cube: } (x+1/x)^3 = x^3 + 3x + 3/x + 1/x^3 = 27. \quad x^3 + 1/x^3 = 27 - 3 \cdot 3 = 18.$$

Subject: Miscellaneous | Topic: Miscellaneous | Difficulty: easy

Q39.

Answer: A

Solution:

$$K_a = \tan^2(45-15) = \tan^2(30) = 1/3.$$

Subject: Geotechnical | Topic: Geotechnical | Difficulty: moderate

Q40.

Answer: C

Solution:

Rolle's: f continuous on $[a,b]$, differentiable on (a,b) , $f(a)=f(b)$. Then $f'(c)=0$ for some c in (a,b) .

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q41.

Answer: A

Solution:

$$P(X \geq 2) = 1 - e^{-4(1+4)} = 1 - 5e^{-4} \approx 0.908.$$

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q42.

Answer: D

Solution:

Perchloric acid (HClO_4) is the strongest among common acids.

Subject: Chemistry | Topic: Chemistry | Difficulty: easy

Q43.

Answer: A

Solution:

$$\text{Area} = \pi r^2. \quad \text{Circumference} = 2\pi r.$$

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q44.

Answer: C

Solution:

$R_A = (L-x)/L$. Linear decreasing from L to 0.

Subject: Structural Analysis | Topic: Structural Analysis | Difficulty: easy

Q45.

Answer: B

Solution:

$\Delta n = 2 - 4 = -2$.

Subject: Chemistry | Topic: Chemistry | Difficulty: difficult

Q46.

Answer: A

Solution:

Row 2 = 2*Row 1, Row 3 = 3*Row 1. All rows linearly dependent. Rank = 1.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: difficult

Q47.

Answer: C

Solution:

Kirchhoff: for K_n , spanning trees = $n^{n-2} = 4^2 = 16$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q48.

Answer: A

Solution:

$FEM = wL \cdot (L/2) = wL^2/2$ (hogging).

Subject: Structural Analysis | Topic: Structural Analysis | Difficulty: easy

Q49.

Answer: A

Solution:

PGF of Poisson(λ): $G(z) = E[z^X] = \exp(\lambda(z-1))$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: easy

Q50.

Answer: A

Solution:

$f(i) = i^2 + 3i + 2 = -1 + 3i$. $\operatorname{Re}(f(i)) = -1$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: easy

Q51.

Answer: A

Solution:

$L\{t^2 f(t)\} = -d/ds(F(s))$. $F(s) = 1/(s+2)$. $-d/ds[1/(s+2)] = 1/(s+2)^2$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q52.

Answer: B

Solution:

$(1+1/n)^{2n} = [(1+1/n)^n]^2 \rightarrow e^2$ as $n \rightarrow \infty$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: easy

Q53.

Answer: B

Solution:

$f(0)=2$, $f(1)=0$, $f(-1)=4$, $f(2)=4$. $f(-2)=-8+6+2=0$. Roots at $x=1$ (double?) and $x=-2$. $f(x)=(x-1)^2(x+2)$. Two distinct real roots.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: easy

Q54.

Answer: A

Solution:

$PI = LL - PL = 60 - 30 = 30$.

Subject: Geotechnical | Topic: Geotechnical | Difficulty: moderate

Q55.

Answer: A

Solution:

Eigenvalues of A^2 are 1, 4, 9. Trace = sum = $1+4+9 = 14$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q56.

Answer: A

Solution:

Homogeneous ODE. Substituting $y=x*v$ and $y(1)=0$ gives $y=0$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q57.

Answer: B

Solution:

$dy/y = dx/x$. $\ln(y) = \ln(x) + C$. $y = C*x$. Wait: $y dx = x dy \Rightarrow dy/y = dx/x \Rightarrow \ln y = \ln x + C \Rightarrow y = C*x$. That's option A. Hmm, let me check: $y dx - x dy = 0 \Rightarrow y dx = x dy \Rightarrow dy/y = dx/x \Rightarrow y = C*x$. Answer: A.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q58.

Answer: A

Solution:

Poisson: mean = lambda = variance. Both = 4.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: easy

Q59.

Answer: C

Solution:

Gauss-Seidel converges if matrix is diagonally dominant. Jacobi needs symmetric positive definite.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: easy

Q60.

Answer: A

Solution:

$|\langle u, v \rangle| \leq \|u\| * \|v\|$. Equality when u and v are linearly dependent.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q61.

Answer: A

Solution:

$L\{e^{(-at)}\} = 1/(s+a)$. Here $a=3$: $1/(s+3)$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q62.

Answer: C

Solution:

$y = A*\cos(2x) + B*\sin(2x)$. $A=1$, $B=0$. $y(\pi/2) = \cos(\pi) = -1$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: difficult

Q63.

Answer: A

Solution:

Eigenvalues are 3 and 1. $[1,0]^T$ is eigenvector for eigenvalue 1. $A^{50}*[1,0]^T = [1,0]^T$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q64.

Answer: B

Solution:

$\text{proj} = ((v \cdot w)/(w \cdot w)) * w = (4/5) * (2, 1, 0) = (8/5, 4/5, 0)$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: easy

Q65.

Answer: B

Solution:

$|x|$ is continuous everywhere. Left derivative at 0 = -1, right = +1. Not differentiable at 0.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: difficult

PAPER BLUEPRINT

Subject Distribution:

Engineering Mathematics: 25 questions, 47 marks
Chemistry: 5 questions, 8 marks
Structural Analysis: 3 questions, 6 marks
Miscellaneous: 3 questions, 6 marks
SOM: 4 questions, 5 marks
Geotechnical: 3 questions, 5 marks
Water Resources Engineering: 4 questions, 4 marks
Civil Engineering: 3 questions, 4 marks
Structural Engineering: 3 questions, 3 marks
General Aptitude: 2 questions, 2 marks
Geotechnical Engineering: 2 questions, 2 marks
Environmental Engineering: 2 questions, 2 marks
Transportation Engineering: 2 questions, 2 marks
Surveying & Geomatics: 2 questions, 2 marks
Construction Materials: 2 questions, 2 marks

Difficulty Distribution:

Easy: 21 | Moderate: 34 | Difficult: 10

Statistics:

Total Questions: 65
Total Marks: 100

This blueprint is generated from the actual question distribution in this paper.